



Shem Yu

ML Engineer, Team Lead

Focus on AI Systems and Agentic Applications



shauns4y@gmail.com, +886 953 240 855

Taiwan, Taipei

Summary

“ Senior Applied AI Engineer specializing in **agent development and multi-agent system architecture**. Experienced in building adaptive task workflows, tool-calling execution engines, repository-integrated validation pipelines, and LLM-based automation. Led enterprise-scale agent projects including DOGI, GAIA, and RKB, deploying autonomous reasoning systems on AWS/Databricks.

Working experience

ML Engineer, MLE Team Lead, AI Dept.

Cathay Financial Holdings • Nov 2023 - Now



- **Led a team of 7** to architect and productionize multi-agent systems, establishing core components adopted across business units.
- **Architected the agent orchestrator layer** for DOGI, GAIA, and RKB, enabling cross-agent communication, workflow orchestration, and tool-calling pipelines.
- **Defined enterprise-wide frameworks** to evaluate agent issues (slow, costly, unstable) and established core agent design patterns adopted across teams.
- **Integrated agent workflows with repository-level validation** and testing routines, supporting automated checks and pipeline-triggered actions.
- **Developed regulation-focused Agent**, improving **F1 from 0.67 to 0.89**, and deployed event-driven batch inference workflows on Databricks.
- **Developed a FinOps agent** for autonomous GPU workload analysis and scaling, reducing GPU cost by 30% and total cloud expenses by 40%.
- Designed and implemented scalable GenAI infrastructure as the agent runtime layer, including the **AI Gateway** and **Guardrails**, reducing service latency by 60%.
- Led cross-team alignment for **GenAI adoption** and standardization of tools, vector DB integrations, and agent workflows.

Senior AI Platform Engineer

Wisers Information Limited • Nov 2020 - May 2022 • 1 year, 8 months



- **Develop standardized ML project templates** (FastAPI, CI/CD, Kubernetes), reducing deployment time from 2 weeks to 3 days, adopted by 30+ projects.
- **Implement NLP-focused Python library** with 70% internal adoption and 90%+ code coverage, improving productivity and reliability for AI developers.
- **Lead automation of documentation pipelines** via Sphinx and CI, enhancing onboarding efficiency and platform usability.
- **Support cross-team AI solution integration**, enabling seamless transition to unified platforms across new and legacy projects.

Data Scientist

TripSaaS • Dec 2019 - Nov 2020 • 1 year

擁樂數據服務

- **Analyze customer order data** using statistical modeling (FP-Growth, K-means) to support marketing and segmentation strategies.
- **Extract and optimize feature patterns** from travel itineraries with CKIP NLP and frequent pattern mining, improving search relevance and user experience.
- **Assist in maintaining ETL pipelines** and data collection, ensuring stable analytics during dynamic business environments.

Intern

Institute for Information Industry, III • May 2017 - Nov 2017 • 6 months



- **Develop full-stack web applications from end to end** for social data analysis, supporting the transformation of raw posts into interactive word clouds and buzzword visualizations.
- **Analyze and extract insights** from social media data, applying NLP and visualization tools to generate actionable reports for industry partners.
- **Facilitate communication and project delivery** between university and III teams, ensuring alignment on deliverables and expectations.

Research assistant

Ming Chuan University - data science Lab. • June 2017 - Jun 2019 • 2 years



- **Lead student research teams** in multiple industry-university collaboration projects, coordinating with external partners and delivering actionable data science solutions.
- **Assist teaching** as a Teaching Assistant for Database Theory, supporting curriculum delivery and mentoring undergraduate students.

Education

Ming Chuan University

Computer Science & Information Engineering • 2017 - 2019

- GPA : 3.82/4
- Master of Computer Science & Information Engineering

- Graduate thesis
 - Yu, H. Y. and Lee, Y. S. "The Portfolio Analysis of the Key Fans for FB fan pages," 2019.
- Presented the research results twice on behalf of the laboratory and won the third prize once

Ming Chuan University

Computer Science & Information Engineering • 2012 - 2016

Skills

Programming & ML Frameworks

Python (expert), R (basic), C/C++ (familiar), Hugging Face, FastAPI, GoogleADK, LangChain, LangGraph, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, SciPy, Matplotlib

MLOps & Deployment

Databricks, MLflow, Docker, CI/CD (Azure DevOps), Kubernetes, AWS SageMaker, AWS Lambda

Cloud & Infra

AWS (EC2, Fargate, Lambda, Route53), Databricks, Distributed Systems, High-Availability Analytics, AI Gateway, Guardrails, MCP Protocol, A2A Protocol

Generative AI & NLP

Agent System Design, Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Prompt Engineering, ASR/TTS, Embedding Databases, Search/Indexing Techniques

Data Engineering

AP Data Layer Design, SQL, NoSQL, Data Pipeline Design, ETL, Data Visualization

Leadership & Communication

Team Leadership, Project Management, Cross-Team Collaboration

Language

- Chinese (Native)
- English (Proficient)